

Goal Prevention 2004

a review of goaltending and team defense
including a study of the quality of
a hockey team's shots allowed

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Introduction

I recently completed an assessment of “Shot Quality” in the NHL for the 2002-03 (“2003”) season (<http://www.HockeyAnalytics.com/Research.htm>). That study revealed that the quality of shots allowed varied significantly from team to team and was not well correlated with the number of shots allowed on goal. The consequence of that study was an improved ability to assess the goal prevention performance of teams and their goaltenders. This paper applies the same methods to the analysis of the 2003-04 (“2004”) season, focusing more on the results than the method.

Shot Quality

In summary, the approach used to assess the quality of shots allowed by a team is:

1. Collect, from NHL game event logs, the relevant data on each shot.
2. Analyze the goal probabilities for each shooting circumstance. In my analysis I separated certain “special shots” from “normal” shots and studied the probability of a goal given the shot type, the shot’s distance and the on-ice situation (power play vs other).
3. Build a model of goal probabilities that relies on the measured circumstance.
4. Apply the model to the shot data for the defensive team in question for the season. For each shot, determine its goal probability.
5. Determine Expected Goals: $EG = \text{the sum of the goal probabilities for each shot.}$
6. Neutralize the variation in the number of shots on goal by calculating Normalized Expected Goals ($NEG = EG \times \text{League Average Shots} / \text{Shots}$)
7. Shot Quality Against ($SQA = NEG / \text{League Average Goals Against.}$)

SQA is defined such that a team SQA of 1.05 means that the team is giving up shots which are 5% more “dangerous” than those of an average team. That team would have, all other things being equal, 5% more goals against than an average team.

There were 69,010 shots on goal in the NHL in 2004 (an average of 2,300 per team). This is down 1.1% from 69,786 (2,326 per team) in the prior season. There were 6,318 goals scored (an average of 211), down 3.3% from 6,530 (218 per team) in 2002-03. Shooting percentage therefore reduced from 9.4% to 9.2%. There are three plausible explanations for this: goaltending got better and/or shooting got worse and/or shot quality was lower. Note that SQA is measured against league average, but there are some

other outputs of the study that can help us with the last possibility. One of these is the average distance of a normal shot. In 2004, this increased by about 0.5 feet suggesting that shot quality did diminish overall this year.

Below are the SQA results for 2004, sorted from best to worst. I have shown the 2003 results and the change so that we can get a better feel for the data.

For the second year in a row, the Devils sit atop the SQA list. Last year's number 2 and 3 teams, Philadelphia and Minnesota, slipped a little in the rankings. But this was because Montreal, Buffalo and Edmonton tightened up defensively.

Buffalo and Montreal were two of the most improved teams in the NHL and part of the story would seem to be shot quality. In the case of Montreal, there was also a dramatic reduction in shots allowed (-334). Edmonton's performance remained about the same notwithstanding the second best improvement in SQA (0.075). SQA was part of the significant defensive improvement, but the Oilers gave up those gains in goal and on offense. Florida escaped the bottom echelon with the largest (0.078) improvement in SQA while other big improvements were NY Islanders (0.072) and Columbus (0.060).

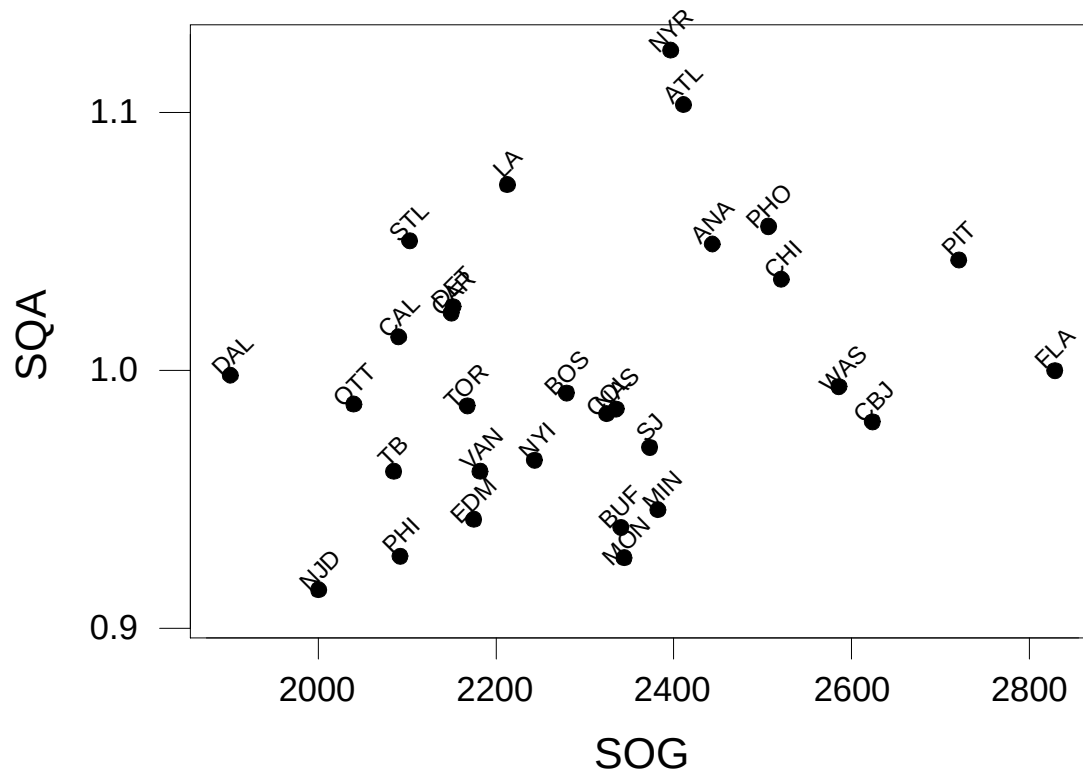
With the exception of Florida, last year's bottom five shuffled the order. Only St. Louis got better, improving to 1.050. Phoenix had one of the largest deteriorations in shot quality (-0.076) to move into the cellar. Other big deteriorations came from Anaheim (-0.077), the Rangers (-0.066), Calgary (-0.060) and Atlanta (-0.059). Calgary had a much better season in spite of this change. The Flames had a complete turnaround in goal. This may have contributed to a change in defensive style. Except for Atlanta, the other teams had worse seasons in 2004.

Shot quality allowed is certainly a function of short handed situations. In 2003, the correlation of these two variables was 45% ($r^2 = 0.20$) and in 2004 the correlation was 57% ($r^2 = 0.32$). But this means that about 75% of the variation in shot quality is still coming from elsewhere. Last year I observed that the correlation of SQA to shots on goal was very weak ($r^2 = .15$). This year it was a bit lower ($r^2 = .12$). This is continued evidence that shots allowed and shot quality allowed are two distinct elements of defensive ability.

SQA Index 2003 vs 2004

Team	2003	2004	Change
NJD	0.915	0.915	0.000
MON	0.982	0.927	0.055
PHI	0.935	0.928	0.007
BUF	0.969	0.939	0.030
EDM	1.018	0.942	0.075
MIN	0.947	0.946	0.000
VAN	1.004	0.961	0.044
TB	0.965	0.961	0.004
NYI	1.037	0.965	0.072
SJ	1.022	0.970	0.052
CBJ	1.041	0.980	0.060
COL	1.018	0.983	0.035
NAS	1.000	0.985	0.015
TOR	0.956	0.986	-0.030
OTT	0.970	0.987	-0.017
BOS	1.027	0.991	0.035
WAS	0.973	0.994	-0.022
DAL	0.972	0.998	-0.025
FLA	1.078	1.000	0.078
CAL	0.953	1.013	-0.060
CAR	1.024	1.022	0.002
DET	0.976	1.025	-0.049
CHI	0.996	1.035	-0.039
PIT	0.996	1.043	-0.047
ANA	0.972	1.049	-0.077
STL	1.087	1.050	0.037
PHO	0.980	1.056	-0.076
LA	1.048	1.072	-0.024
ATL	1.045	1.103	-0.059
NYR	1.057	1.124	-0.066

Do teams need to sacrifice shot quality to reduce shots on goal? The answer is in the chart below:



The answer is clearly no. New Jersey excels at both. Philadelphia excels at both. However, poorer defensive teams may see this as a choice. They tend to form a NW-SE line through the upper right quadrant of the data.

Goaltending

The value of SQA is that it gives us a better understanding of goal prevention. Before consideration of shot quality goal prevention looked like a simple model:

$$GA = SOG \times (1 - SV)$$

where
 SOG = Shots on Goal
 SV = Save Percentage

We knew that Shots on Goal was basically a defensive responsibility, so we have historically attributed it to the team. We knew that Save Percentage was more a goaltender statistic than a team statistic, so we have historically attributed it to goaltenders. But we are now in a position to use a better model:

$$GA = SQA \times SOG \times (1 - SQNSV)$$

where

$$SQNSV = 1 - (1 - SV) / SQA$$

the Shot Quality Neutral

Save Percentage

In this model we attribute both SQA and SOG to the defense and SQNSV to goaltending. Clearly SQNSV is a better measure of the goaltender's contribution to team success than is SV. You can think of it as the save percentage one would expect with no variation in shot quality from team to team.

To the right are the SQNSV calculations for 2004 with a comparison to raw Save Percentage and to SQNSV for 2003.

When comparing the two years you should note that, as goal scoring was down in 2004, save percentages are up versus 2003.

The average save percentage increased by about 0.002.

Florida, nosed out by Minnesota for top goaltending honours in 2003, claimed the top spot in 2004. Notice how the math works for the Panthers. As their $SQA = 1.000$, their SQNSV is equal to SV. Also notice that the math lets a little bit of the air out of the save percentages for each of San Jose and Minnesota. Those teams did a better job minimizing shot quality than did the Panthers.

Shot Quality Neutral Save Percentage 2003 vs 2004

Team	2003		2004				
	SQNSV	Rank	SQA	SV	Rank	SQNSV	Rank
FLA	0.919	2	1.000	0.922	3	0.922	1
SJ	0.902	21	0.970	0.923	2	0.920	2
MIN	0.919	1	0.946	0.923	1	0.919	3
ANA	0.916	4	1.049	0.913	9	0.917	4
CAL	0.892	30	1.013	0.916	7	0.917	5
BOS	0.901	24	0.991	0.918	6	0.917	6
DET	0.912	7	1.025	0.912	10	0.914	7
COL	0.918	3	0.983	0.915	8	0.913	8
MON	0.911	8	0.927	0.918	4	0.912	9
NJD	0.906	14	0.915	0.918	5	0.910	10
STL	0.900	25	1.050	0.906	20	0.910	11
ATL	0.895	29	1.103	0.899	27	0.909	12
LA	0.901	23	1.072	0.902	26	0.909	13
DAL	0.916	5	0.998	0.908	14	0.908	14
VAN	0.905	16	0.961	0.911	11	0.907	15
PHO	0.905	17	1.056	0.902	24	0.907	16
CBJ	0.904	18	0.980	0.909	13	0.907	17
NYR	0.910	10	1.124	0.896	29	0.907	18
OTT	0.908	13	0.987	0.907	16	0.906	19
NAS	0.909	11	0.985	0.907	17	0.906	20
CAR	0.900	26	1.022	0.903	23	0.905	21
TOR	0.910	9	0.986	0.906	19	0.905	22
PHI	0.912	6	0.928	0.911	12	0.904	23
TB	0.905	15	0.961	0.908	15	0.904	24
NYI	0.904	19	0.965	0.906	18	0.903	25
WAS	0.908	12	0.994	0.902	25	0.902	26
CHI	0.903	20	1.035	0.897	28	0.901	27
BUF	0.902	22	0.939	0.906	21	0.899	28
EDM	0.899	27	0.942	0.904	22	0.899	29
PIT	0.899	28	1.043	0.889	30	0.893	30

Big Changes in Goaltending:

Calgary (SQNSV +25 points):

		2004					2003			
Player	Team	GS	SOG	SV	SQNSV	Chg?	GS	SOG	SV	SQNSV
Roman Turek	CAL	18	463	0.914	0.915		65	1679	0.902	0.897
Jamie McLennan	CAL	24	587	0.910	0.911		17	537	0.892	0.887
Miikka Kiprusoff	CAL	38	966	0.933	0.934					
Dany Sabourin	CAL	2	66	0.848	0.850					

Ranked DFL (.892) in 2003 but climbed to 5th (.917) in 2004. SQNSV improved 25 points. Last season Turek got 65 starts and McLennan got 17 starts. Both had awful SQNSV numbers. This season Kiprusoff (.934) gets 38 starts, McLennan (.911) gets 24, Turek (.915) gets 18 and Sabourin (.850) gets 2. Clearly the biggest win in Calgary was acquiring and playing Kiprusoff. But McLennan (traded late in the season) and Turek played much better (green in the “Chg?” column means a statistically significant improvement, red means a significant deterioration). Kiprusoff was a different man in the different uniform (see San Jose) and the biggest goaltending story of the year. He contributed way more than expected in Calgary, only getting installed as the main man in December.

Calgary also had a big deterioration in shot quality. It is possible that part of the big “improvement” in Calgary’s goaltending is due to the inadequacies of SQA. It is possible that the Flames, confident that Kipper would get the job done, traded off shot quality for a better transition game. There are so many synergies in hockey ...

San Jose (+18):

		2004					2003			
Player	Team	GS	SOG	SV	SQNSV	Chg?	GS	SOG	SV	SQNSV
Evgeni Nabakov	SJ	58	1610	0.921	0.919		55	1561	0.906	0.909
Miikka Kiprusoff	SJ						18	537	0.879	0.882
Vesa Toskala	SJ	24	760	0.930	0.928		9	287	0.927	0.928

Ranked 21st in goaltending (.902 SQNSV) in 2003 but 2nd in 2004 (.920). In 2003 it was 55 starts for Nabakov (.909), 18 for Kiprusoff (.882) and 9 for Toskala (.928). San Jose had an SQA of 1.022. In 2004, it was 58 starts for Nabakov (.919 SQNSV) and 24 for Toskala (.928). Toskala was the same, strong goaltender and picked up Kiprusoff’s workload. Nabakov 10 point improvement was significant (about 2 goals against for every 7 games played), **but more of the team improvement was from playing Toskala over Kiprusoff**. And the improvement would have been bigger had they played Toskala over Nabakov. Toskala was statistically the better of the two in 2004. Unless he was getting the softer starts, he was the better man.

Boston (+16):

Except for Gerry Cheevers, I cannot think of a single goaltending icon to spend any amount of time in a Bruins uniform since the 1930s. Maybe that is changing. Certainly Raycroft is the rookie of the year.

		2004					2003			
Player	Team	GS	SOG	SV	SQNSV	Chg?	GS	SOG	SV	SQNSV
Steve Shields	BOS						34	930	0.896	0.898
John Grahame	BOS						22	625	0.902	0.905
Jeff Hackett	BOS						18	500	0.894	0.897
Andrew Raycroft	BOS	55	1586	0.926	0.926		5	146	0.918	0.920
Tim Thomas	BOS						3	118	0.907	0.909
Felix Potvin	BOS	27	690	0.903	0.902					

The Bruins SQNSV improved 16 points and they climbed from 24th (.901) in 2003 to 6th (.917) in 2004. In 2003 Boston installed a revolving door in the net: Shields (34 starts, .898 SQNSV), Grahame (22, .905), Hackett (18, .897) and Thomas (3, .899) all had a shot at it. Only Raycroft (5 starts) posted a strong SQNSV (.920). This season Raycroft (.926) had 55 starts while Potvin (.902) had 27. The Cat was not much better than the average of last year's menagerie. Note that his performance in Boston was similar to that in LA. Raycroft was a bit better than last year, **but the big difference was playing him a lot!**

Atlanta (+14):

		2004					2003			
Player	Team	GS	SOG	SV	SQNSV	Chg?	GS	SOG	SV	SQNSV
Pasi Nurminen	ATL	62	1792	0.903	0.913		46	1452	0.906	0.910
Byron Dafoe	ATL	16	499	0.898	0.907		17	472	0.862	0.868
Kari Lehtonen	ATL	4	106	0.953	0.957					
Milan Hnlicka	ATL						17	605	0.893	0.897
Frederic Cassivi	ATL						2	58	0.810	0.818

The Thrashers' goaltending moved from 29th to 12th (SQNSV improved from .895 to .909), but it may not have been noticeable with the deterioration in shot quality. In 2003 split the goaltending starts as follows: Nurminen 46 (SQNSV .910), Dafoe 17 (.868), Hnlicka 17 (.897) and Cassivi 2 (.818). This season it was Nurminen 62 (.913), Dafoe 16 (.907) and Lehtonen 4 (.957).

In 2004, Nurminen was again was the main man. **He was basically the same goaltender** (remember the league average save percentage was up by 2 points) but he played a lot more. The improvement in Atlanta was from (a) replacing Hnlicka's starts with Nurminen and (b) a big improvement from Dafoe.

St. Louis (+10):

The Blues goaltending in 2003 was hardly stellar. An SQA of 1.087 made it look even worse. In 2004 goaltending improved and it is pretty easy to see why. Chris Osgood delivered a SQNSV of .915 in 65 starts. In 2003 with both St. Louis and the Islanders,

Osgood had a SQNSV of .897. This is a huge improvement. I didn't think he had it in him. Johnson was the same goalie. Divis was never going to repeat 2003.

Player	Team	2004					2003			
		GS	SOG	SV	SQNSV	Chg?	GS	SOG	SV	SQNSV
Brent Johnson	STL	8	203	0.901	0.906		35	844	0.900	0.908
Fred Brathwaite	STL						23	631	0.883	0.892
Chris Osgood	STL	65	1604	0.910	0.915		9	241	0.888	0.897
Curtis Sanford	STL						7	148	0.912	0.919
Tom Barrasso	STL						6	132	0.879	0.889
Reinhard Divis	STL	9	291	0.900	0.905		2	34	0.971	0.973
Cody Rudkowsky	STL						0	10	1.000	1.000

Philadelphia (-8) vs Los Angeles (+8):

Bobby Clarke doesn't want to read this but ... Philadelphia's goaltending was the mirror image of Boston's, dropping from 6th to 23rd. The Flyers' SQNSV dropped 8 points from .912 to .904 while the Kings' SQNSV increased 8 points from .901 to .909 (and the Kings' goaltending rank moved from 23rd to 13th). What caused these changes? The Flyers dispatched Roman Cechmanek to LA at the end of 2003.

Player	Team	2004					2003			
		GS	SOG	SV	SQNSV	Chg?	GS	SOG	SV	SQNSV
Roman Cechmanek	PHI						57	1368	0.925	0.920
Robert Esche	PHI	40	932	0.915	0.909		25	647	0.907	0.901
Jeff Hackett	PHI	26	684	0.905	0.898					
Sean Burke	PHI	13	389	0.910	0.903					
Antero Niittymäki	PHI	3	77	0.961	0.958					
Neil Little	PHI	0	8	0.750	0.731					

In 2003, Philly gave 57 starts to Cechmanek (.920) and 25 to Esche (.901). In 2004 Esche (.909) got 40 starts, Hackett (.898) got 26, Burke (.903) got 13 and Niittymäki (.958) got 3. Esche got better in 2004. But since he never caught up to Cechmanek's SQNSV, playing him more hurt the team.

Player	Team	2004					2003			
		GS	SOG	SV	SQNSV	Chg?	GS	SOG	SV	SQNSV
Felix Potvin	LA						41	987	0.894	0.898
Jamie Storr	LA						33	904	0.905	0.909
Cristobal Huet	LA	33	961	0.907	0.914		8	241	0.913	0.917
Roman Cechmanek	LA	48	1198	0.906	0.912					
Milan Hnilicka	LA	1	42	0.881	0.889					
Mathieu Chouinard	LA	0	2	1.000	1.000					

LA went from Potvin (41 starts, .898 SQNSV), Storr (33, .909) and Huot (8, .917) in 2003 to Cechmanek (48, .912), Hout (33, .914) and Hnilicka (1, .889) in 2004.

On the surface it looks like Cechmanek (.906 SV) was weak in LA and Esche (.915 SV) rose to the occasion in Philly. But LA had an SQA of 1.072 and Philadelphia has an SQA of .928. I will stick with this analysis. It says that Cechmanek did not play as well as he did in Philly, but **the Flyers hurt themselves in goal with the trade.**

Dallas (-8):

The Stars goals against went from 169 to 175 in spite of reducing shots on goal from 2073 to 1902 and in spite of league scoring going down in 2004. Shot quality deteriorated slightly (.972 to .998). This would account for part of the story. But the goaltending was off as well.

		2004					2003			
Player	Team	GS	SOG	SV	SQNSV	Chg?	GS	SOG	SV	SQNSV
Marty Turco	DAL	72	1648	0.913	0.912		52	1359	0.932	0.930
Ron Tugnutt	DAL	9	220	0.900	0.900		29	672	0.896	0.893
Corey Hirsch	DAL						1	39	0.897	0.895
Dan Ellis	DAL	1	28	0.893	0.893					

Two things happened to Marty Turco in 2004: he played more and he played worse. He had an awesome season in 2003. Perhaps that was as good as it gets. Perhaps Belfour was the keeper.

Same Goaltending:**Florida (+3):**

		2004					2003			
Player	Team	GS	SOG	SV	SQNSV	Chg?	GS	SOG	SV	SQNSV
Roberto Luongo	FLA	72	2475	0.931	0.931		62	2011	0.918	0.924
Jani Hurme	FLA						20	707	0.907	0.913
Steve Shields	FLA	10	346	0.879	0.879					

Although Florida had a dramatic improvement in its shot quality, the goaltending metric (SQNSV) hardly moved. Better defense, same goaltending? Not quite. Luongo was even better in 2004 (SQNSV went from .924 to .931) AND started in 10 additional games. But Shields (.879, 10 starts) was significantly worse than Hurme (.913, 20 starts). But guess what? **Because the defense improved, people started to notice Luongo.**

Other than Florida, the top 5 teams all had very capable backups, in some cases playing a great deal. The Panthers were the exception to the rule and just kept trotting out Luongo.

Anaheim (+1):

		2004					2003			
Player	Team	GS	SOG	SV	SQNSV	Chg?	GS	SOG	SV	SQNSV
Jean-Sebastien Giguere	ANA	52	1623	0.914	0.918		64	1820	0.920	0.918
Martin Gerber	ANA	29	785	0.918	0.922		18	548	0.929	0.927
Ilya Bryzgalov	ANA	1	28	0.929	0.932					

Anaheim had the largest deterioration in shot quality allowed (SQA went from .972 to 1.049) and SQNSV hardly moved. **Same goaltending, worse defense!** Almost. In Anaheim they were calling Giguere a sieve. Yet he had a SQNSV of .918 in both 2003 and 2004. In fact, team defense got worse and last year's Conn Smythe Trophy winner got an early start to the golf season

Martin Gerber played more in 2004 and played better than Giguere (for the second year in a row) but had a slightly lower SQNSV in 2004 (.922 vs .927). It seems to me that he has earned a shot at a starting job (any direct flights to Ottawa?).

Minnesota (-):

		2004					2003			
Player	Team	GS	SOG	SV	SQNSV	Chg?	GS	SOG	SV	SQNSV
Dwayne Roloson	MIN	46	1323	0.933	0.929		47	1334	0.927	0.922
Manny Fernandez	MIN	36	1056	0.915	0.910		34	972	0.924	0.920
Dieter Kochan	MIN						1	28	0.821	0.811

Nothing much changed for the Wild: Not shots on goal (2383 in 2003 to 2335 in 2004). Not shot quality (SQA: .947 to .946). And not goaltending (SQNSV: .919 to .919). Goaltending duties were split in about the same fashion.

Roloson got even better (.922 to .929). When will he get credit for his performance? But Fernandez got significantly worse (.920 to .910).

Toronto (-5) and Colorado (-5):

		2004					2003			
Player	Team	GS	SOG	SV	SQNSV	Chg?	GS	SOG	SV	SQNSV
Ed Belfour	TOR	58	1483	0.918	0.917		62	1816	0.922	0.919
Trevor Kidd	TOR	14	388	0.876	0.875		19	565	0.896	0.891
Mikael Tellqvist	TOR	10	293	0.894	0.893		1	38	0.895	0.890

Toronto's goaltending was off by 5 points, but it really wasn't Ed Belfour. Yes, Belfour's SQNSV went down by 2 points. But more of the story is at the end of the bench. Belfour played a little less (bad back) and it became clearer that **this team really needs a competent back up.**

		2004					2003			
Player	Team	GS	SOG	SV	SQNSV	Chg?	GS	SOG	SV	SQNSV
Patrick Roy	COL						63	1723	0.920	0.922
David Aebischer	COL	61	1703	0.924	0.923		19	593	0.916	0.917
Philippe Sauve	COL	16	479	0.896	0.894					
Tommy Salo	COL	5	136	0.912	0.910					

The story was similar in Colorado. At the beginning of the season everyone wondered if Aebischer could fill Roy's shoes. This analysis says that he did (and that he is a pretty good goaltender). **The problem in Colorado was that Sauve couldn't fill Aebischer's shoes.** Salo was added at the trade deadline. But he can't fill those shoes either. The result was that Colorado's goaltending was off by 5 points.

Detroit (+2):

Here is an interesting analysis. Detroit's goaltending got better by 2 points in 2004, in line with the league average improvement. Joseph was basically the same goaltender from season to season. The Dominator came back, played a little, developed a sore groin

and took a long rest. But he had the same numbers as Joseph. And Legace was basically the same goaltender he was in 2003. So what happened? How did Detroit's goaltending improve? **The NHL's best back up goaltender got more work!** Has he finally earned the number one job?

		2004					2003			
Player	Team	GS	SOG	SV	SQNSV	Chg?	GS	SOG	SV	SQNSV
Curtis Joseph	DET	30	744	0.909	0.911		60	1676	0.912	0.910
Manny Legace	DET	36	1019	0.920	0.921		22	681	0.925	0.923
Dominik Hasek	DET	14	324	0.907	0.910					
Marc Lamothe	DET	2	58	0.948	0.950					

New Jersey (+4):

		2004					2003			
Player	Team	GS	SOG	SV	SQNSV	Chg?	GS	SOG	SV	SQNSV
Martin Brodeur	NJD	75	1845	0.917	0.909		73	1706	0.914	0.906
Corey Schwab	NJD	3	68	0.971	0.968		9	223	0.933	0.926
Scott Clemmensen	NJD	4	84	0.952	0.948					

Brodeur was basically the same goaltender in 2004. He still plays behind the league's best defense and he still doesn't deserve the Vezina.

Montreal (+1):

		2004					2003			
Player	Team	GS	SOG	SV	SQNSV	Chg?	GS	SOG	SV	SQNSV
Jose Theodore	MON	66	1860	0.919	0.913		57	1797	0.908	0.907
Jeff Hackett	MON						17	606	0.926	0.924
Mathieu Garon	MON	16	480	0.921	0.915		8	267	0.940	0.939

Montreal's goals against dropped from 234 to 192. It had to be goaltending, right? Montreal had a huge reduction in shots on goal and a big improvement in shot quality. Theodore bounced back from a disappointing season, but his improvement was not "statistically significant". Offsetting this improvement was the decline in back up goaltending. **The big reduction in goals against came from a huge improvement in team defense.**

Vancouver (+2):

		2004					2003			
Player	Team	GS	SOG	SV	SQNSV	Chg?	GS	SOG	SV	SQNSV
Dan Cloutier	VAN	60	1554	0.914	0.910		57	1477	0.908	0.908
Peter Skudra	VAN						20	522	0.897	0.897
Alex Auld	VAN	5	168	0.929	0.926		5	165	0.939	0.940
Tyler Moss	VAN						0	14	0.929	0.929
Johan Hedberg	VAN	17	459	0.900	0.896					

Cloutier was no better in 2004 (league average change) but played a bit more. Hedberg couldn't play in Pittsburgh and was a liability in Vancouver.

Phoenix (+2):

Player	Team	2004				Chg?	2003			
		GS	SOG	SV	SQNSV		GS	SOG	SV	SQNSV
Brian Boucher	PHO	39	1150	0.906	0.911		40	1210	0.894	0.892
Sean Burke	PHO	29	913	0.908	0.913		22	632	0.930	0.929
Zac Bierk	PHO	4	108	0.889	0.895		15	471	0.932	0.931
Patrick DesRochers	PHO						3	88	0.875	0.872
Jean-Marc Pelletier	PHO	2	84	0.857	0.865		2	48	0.875	0.872
Brent Johnson	PHO	8	243	0.914	0.918					

Boucher was way better in 2004, **but a huge deterioration in shot quality made this less apparent.** Burke, coming off a stellar season, was much worse (and got worse in Philly).

Columbus (+3):

Player	Team	2004				Chg?	2003			
		GS	SOG	SV	SQNSV		GS	SOG	SV	SQNSV
Marc Denis	CBJ	63	1970	0.918	0.916		77	2404	0.903	0.907
Jean-Francois Labbe	CBJ						5	233	0.884	0.889
Fred Brathwaite	CBJ	18	574	0.897	0.895					
Pascal Leclaire	CBJ	2	69	0.899	0.896					

Denis got significantly better. Although he played less and the back ups were weak, the Blue Jackets improve their goaltending by 3 points. Brathwaite couldn't cut it in St. Louis (SQNSV .892) and he couldn't cut it behind better shot quality.

NY Rangers (-3):

Player	Team	2004				Chg?	2003			
		GS	SOG	SV	SQNSV		GS	SOG	SV	SQNSV
Mike Dunham	NYR	54	1522	0.896	0.907		43	1229	0.924	0.928
Dan Blackburn	NYR						27	842	0.890	0.896
Mike Richter	NYR						12	329	0.897	0.902
Johan Holmqvist	NYR						0	18	0.889	0.895
Jussi Markkanen	NYR	19	611	0.913	0.923					
Jamie McLennan	NYR	4	97	0.876	0.890					
Jason Labarbera	NYR	3	91	0.824	0.844					
Stephen Valiquette	NYR	2	71	0.915	0.925					

Dunham had a great season in 2003 that went largely unnoticed. **His performance was way off in 2004 but it looked even worse as the Rangers ranked last in shot quality allowed** (SQA 1.124). Markkanen played very well in relief, but Sather traded him ...

Ottawa (+2):

Player	Team	2004				Chg?	2003			
		GS	SOG	SV	SQNSV		GS	SOG	SV	SQNSV
Patrick Lalime	OTT	55	1334	0.905	0.904		65	1591	0.911	0.908
Martin Prusek	OTT	26	651	0.917	0.916		16	415	0.911	0.908
Ray Emery	OTT	1	52	0.904	0.903		1	26	0.923	0.921

Ottawa's improvement in goal (which only allowed them to keep pace with the league) came from the bench. Lalime had his SQNSV slip from .908 to .904. But he started 10 fewer games and Prusek was better. In fact, his SQNSV of .916 was **statistically** better than Lalime's. **When will Ottawa realize that Lalime, who has the numbers of a competent back up, will not take them to the Stanley Cup?**

Nashville (-3)

		2004					2003			
Player	Team	GS	SOG	SV	SQNSV	Chg?	GS	SOG	SV	SQNSV
Tomas Vokoun	NAS	73	1958	0.909	0.908		68	1771	0.918	0.918
Mike Dunham	NAS						13	397	0.892	0.892
Jan Lasak	NAS						1	39	0.872	0.872
Brian Finley	NAS						0	13	0.769	0.769
Wade Flaherty	NAS						0	27	0.852	0.852
Chris Mason	NAS	9	365	0.926	0.925					

Vokoun had a significantly worse season. But the weak performance of the back ups in the 14 games started in 2003 was mitigated by (1) 5 additional starts for Vokoun and Mason's strong work in limited action.

Carolina (+5)

		2004					2003			
Player	Team	GS	SOG	SV	SQNSV	Chg?	GS	SOG	SV	SQNSV
Kevin Weekes	CAR	65	1652	0.912	0.914		49	1438	0.912	0.914
Arturs Irbe	CAR	9	228	0.899	0.901		31	816	0.877	0.880
Patrick DesRochers	CAR						2	71	0.901	0.904
Jamie Storr	CAR	8	262	0.878	0.881					

Carolina's improved SQNSV (.900 to .905) was mainly due to playing Weekes considerably more. But Irbe rescued his sagging career, at least enough to be seen as a competent back up.

Tampa Bay (-1)

		2004					2003			
Player	Team	GS	SOG	SV	SQNSV	Chg?	GS	SOG	SV	SQNSV
Nikolai Khabibulin	TB	54	1414	0.910	0.907		64	1760	0.911	0.908
John Grahame	TB	28	664	0.913	0.909		15	424	0.920	0.917
Kevin Hodson	TB						3	101	0.881	0.877
Evgeny Konstantinov	TB						0	6	0.833	0.827

Khabibulin was basically the same goaltender. He gave up some playing time to Grahame who, although his SQNSV deteriorated in 2004, had slightly better numbers. **Is this Stanley Cup goaltending?**

NY Islanders (-1)

Snow stumbled in 2004. DiPietro picked up Osgood's workload and emerged as the number one goalie. But he still isn't Luongo.

		2004					2003			
Player	Team	GS	SOG	SV	SQNSV	Chg?	GS	SOG	SV	SQNSV
Garth Snow	NYI	34	932	0.899	0.896		37	1120	0.918	0.921
Chris Osgood	NYI						36	912	0.894	0.897
Rick DiPietro	NYI	47	1261	0.911	0.908		9	273	0.894	0.898
Wade Dubielewicz	NYI	1	50	0.940	0.938					

Washington (-6)

		2004					2003			
Player	Team	GS	SOG	SV	SQNSV	Chg?	GS	SOG	SV	SQNSV
Olaf Kolzig	WAS	63	1958	0.908	0.908		65	1925	0.919	0.917
Sebastien Charpentier	WAS	6	168	0.875	0.874		12	426	0.906	0.903
Craig Billington	WAS						5	96	0.823	0.818
Maxime Ouellet	WAS	6	210	0.910	0.909					
Matt Yeats	WAS	4	141	0.908	0.907					
Rastislav Stana	WAS	3	100	0.890	0.889					

I guess it was hard to focus in Washington with the fire sale taking place. Kolzig's deterioration was significant. His numbers looked much like Vokoun. Apparently he held out for a trade to Colorado. I don't know why he would want to compete with Aebischer.

Chicago (-2)

		2004					2003			
Player	Team	GS	SOG	SV	SQNSV	Chg?	GS	SOG	SV	SQNSV
Jocelyn Thibault	CHI	14	450	0.913	0.916		62	1690	0.915	0.914
Steve Passmore	CHI	8	221	0.896	0.899		9	284	0.866	0.866
Michael Leighton	CHI	33	987	0.900	0.903		6	241	0.913	0.912
Craig Andersson	CHI	19	602	0.905	0.909		5	125	0.856	0.855
Adam Munro	CHI	7	217	0.880	0.884					
Matt Underhill	CHI	1	33	0.879	0.883					

The story of the Blackhawks was Thibault's playing time. The backups were better in 2004, but nobody else could rise to his level.

Buffalo (-3)

		2004					2003			
Player	Team	GS	SOG	SV	SQNSV	Chg?	GS	SOG	SV	SQNSV
Martin Biron	BUF	49	1442	0.913	0.908		53	1468	0.908	0.905
Mika Noronen	BUF	30	821	0.906	0.900		14	411	0.912	0.910
Ryan Miller	BUF	3	73	0.795	0.781		15	410	0.902	0.899

There was no real change here. Buffalo had a noticeable improvement in shot quality, so you need to let some of the air out of Biron's improved save percentage. Nevertheless, 2004 was a bit better for him. Noronen won the back up job in 2003 and picked up Miller's work for 2004. Then he produced a result that was the average of the two.

Edmonton (-)

Player	Team	2004				Chg?	2003			
		GS	SOG	SV	SQNSV		GS	SOG	SV	SQNSV
Tommy Salo	EDM	41	1024	0.896	0.889		65	1708	0.899	0.901
Jussi Markkanen	EDM	6	182	0.934	0.930		17	533	0.904	0.906
Ty Conklin	EDM	35	959	0.912	0.907					
Stephen Valiquette	EDM	0	7	0.714	0.697					

Salo had a weak season in 2003 and a weaker one in 2004. He might yet salvage his career as a backup. His playing time was reduced in favour of Conklin, who brought Edmonton's numbers back to 2003 levels. However, Markkanen looks like the future for the Oilers.

Pittsburgh (-6)

Player	Team	2004				Chg?	2003			
		GS	SOG	SV	SQNSV		GS	SOG	SV	SQNSV
Johan Hedberg	PIT						39	1197	0.895	0.894
Sebastien Caron	PIT	37	1179	0.883	0.888		23	741	0.916	0.916
Jean-Sebastien Aubin	PIT	17	574	0.908	0.911		20	589	0.900	0.899
Marc-Andre Fleury	PIT	20	675	0.896	0.901					
Andy Chiodo	PIT	8	260	0.892	0.897					
Martin Brochu	PIT	0	19	0.947	0.950					

Pittsburgh had a sad defense in 2004, so you have to pump some air into the goaltending numbers. Having said that, only Aubin played well. He moved his save percentage up from .900 to .908 behind deteriorating shot quality. Fleury started the season with a flurry. He was shipped back to Cape Breton after 20 sometimes brilliant starts. Caron faded from a pretty strong season in 2003.

Individual Goaltending Leaders

To the right are the top 10 goaltenders in 2004. This list is ranked by SQNSV but only includes goalies who started in 50% or more of their team's games. You could argue a fair bit about that cut off. Today's prima-goalies tend to get at least 50 starts, but that depends on context and injuries.

If you expand this list a bit you get into the Kiprusoff

Individual SQNSV Leaders 2004					
Player	Team	GS	SOG	SV	SQNSV
Roberto Luongo	FLA	72	2475	0.931	0.931
Dwayne Roloson	MIN	46	1323	0.933	0.929
Andrew Raycroft	BOS	55	1586	0.926	0.926
David Aebischer	COL	61	1703	0.924	0.923
Evgeni Nabokov	SJ	58	1610	0.921	0.919
Jean-Sebastien Giguere	ANA	52	1623	0.914	0.918
Ed Belfour	TOR	58	1483	0.918	0.917
Marc Denis	CBJ	63	1970	0.918	0.916
Chris Osgood	STL	65	1604	0.910	0.915
Special Consideration					
Player	Team	GS	SOG	SV	SQNSV
Miiikka Kiprusoff	CAL	38	966	0.933	0.934
Manny Legace	DET	36	1019	0.920	0.921

debate. Here is a guy who (a) took a while to win the job and (b) was out for a month. Is it fair to exclude him from the list? Is it fair to include him on the list? And what about the Vezina?

One way of looking at this is to look at confidence intervals. I am 95% confident that Roberto Luongo has a “true” normalized save percentage of between .928 and .933 (on a team with average shot quality like the Panthers). I am 95% confident that Mikka Kiprusoff has a true normalized save percentage of between .930 and .937 (on a team with average shot quality like the Flames). There is an overlap in those ranges, so I cannot be confident that either was the better goalie in 2004. In fact, we cannot dismiss Roloson from the analysis since his confidence intervals also intersect those of Luongo and Kiprusoff

My tie breaker is therefore the significantly higher workload for Luongo. My Player Contribution system for the evaluation of player performance (see *Player Contribution* at <http://www.HockeyAnalytics.com/Research.htm>) takes workload into consideration and Luongo has a PC score of 348 versus Kiprusoff’s 143. A PC score of 348 suggests that Florida would have been about 35 points worse off in the standings if they had played a “marginal” goaltender in place of Luongo. I have been doing some historical work on PC scores for goaltenders and this is, by far, the highest PC score I have come across. Below are the top 10 PC scores for goaltenders in 2004:

Player	Team	PC	PCG	GS	MIN	GA	GAA	SA	SV%	SQNSV
Roberto Luongo	FLA	348	345	72	4252	172	2.43	2475	0.931	0.931
Andrew Raycroft	BOS	202	199	55	3420	117	2.05	1586	0.926	0.926
David Aebischer	COL	174	174	61	3703	129	2.09	1703	0.924	0.923
Dwayne Roloson	MIN	162	163	46	2847	89	1.88	1323	0.933	0.929
Marc Denis	CBJ	156	156	63	3796	162	2.56	1970	0.918	0.916
Evgeni Nabokov	SJ	144	147	58	3456	127	2.20	1610	0.921	0.919
Miikka Kiprusoff	CAL	143	142	38	2301	65	1.69	966	0.933	0.934
Jean-Sebastien Giguere	ANA	136	135	52	3210	140	2.62	1623	0.914	0.918
Jose Theodore	MON	129	127	66	3961	150	2.27	1860	0.919	0.913
Chris Osgood	STL	129	132	65	3861	144	2.24	1604	0.910	0.915

The Vezina Trophy:

The Vezina goes to the goalkeeper adjudged to be “the best at this position as voted by the general managers of all NHL clubs”. What does it say about NHL GMs that Brodeur, again, does not belong on the list of finalists? Behind what is certainly the NHL’s best defense, his goals against average and save percentage were not impressive. I don’t get it.

Luongo was, without question, the most valuable goalie to his team, but the trophy is for “best at his position”. Kiprusoff played as well (actually better, but not statistically so). Calgary’s goaltending (not just Kiprusoff) was totally turned around in 2004. This got a lot of media attention. Florida continues to be an awful team (just wait for a Luongo injury) and gets ignored. I think that impact counts and I don’t think it is close. Luongo is my Vezina man.

Team Defense

The goal prevention formula

$$GA = SQA \times SOG \times (1 - SQNSV)$$

gives us additional insight into goaltending, but it also gives us another view of defense. If we attribute only SQNSV to goaltending, then a single metric that captures all information about defense is Shot Quality Neutral Shots on Goal:

$$SQNSOG = SQA \times SOG$$

SQNSOG is a bit like “wind chill” for goaltenders – a team allows 30 shots on goal but, to the goaltender, it feels like 35. To the right are the 2004 rankings of team defense based on this metric. I have shown SQNSOG per game as shots on goal per game is discussed more frequently than season shots.

The top of this list did not change much. Tampa Bay moved from 8th to 4th while Calgary slipped from 5th to 8th.

The bottom of the list changed a bit more. Florida gave up big improvements in shot quality by allowing more shots on goal. The next effect was nearly neutral. Atlanta went in the opposite direction, also with neutral effect. The Rangers allowed a similar number of shots on goal but shot quality went south. Phoenix got worse in both dimensions.

The biggest improvements in team defense rankings came from Montreal (+16) and the Islanders (+12). Both of these teams had big improvements in shot quality. Montreal had a huge improvement in shots allowed. New York's improvement was more modest. Other big improvements came from Carolina (+8, mainly shots), Edmonton (+7, mainly shot quality) and Toronto (+7 even though SQA deteriorated).

The biggest deteriorations in team defense were from Los Angeles (-10), Minnesota (-9), Chicago (-8), Anaheim (-7) and Nashville (-7). All of these teams allowed more shots. The Wild had stable shot quality. Only Nashville improved SQA.

**Shot Quality Neutral Shots on Goal
2003 vs 2004**

	2003		2004	
Team	SQNSOG/g	2003	SQNSOG/g	Rank
NJD	21.6	1	22.3	1
DAL	24.6	4	23.1	2
PHI	23.0	2	23.7	3
TB	27.0	8	24.4	4
OTT	24.0	3	24.5	5
EDM	27.9	13	25.0	6
VAN	26.8	6	25.6	7
CAL	25.8	5	25.8	8
TOR	28.3	16	26.1	9
NYI	29.3	22	26.4	10
MON	32.1	27	26.5	11
BUF	27.1	9	26.8	12
CAR	29.1	21	26.8	13
DET	28.1	14	26.9	14
STL	27.1	10	26.9	15
MIN	27.0	7	27.5	16
BOS	29.1	19	27.6	17
COL	28.8	18	27.9	18
NAS	27.5	12	28.0	19
SJ	29.8	24	28.1	20
LA	27.4	11	28.9	21
ANA	28.2	15	31.3	22
CBJ	33.5	29	31.3	23
WAS	29.1	20	31.4	24
CHI	28.5	17	31.8	25
PHO	29.4	23	32.3	26
ATL	33.0	28	32.4	27
NYR	31.3	26	32.8	28
FLA	35.8	30	34.5	29
PIT	30.8	25	34.6	30

Goal Prevention Indices

If you express team goals against as a percentage of league average goals against, you get team defense expressed in terms of league averages:

$$\begin{aligned} GA / GA_{<} &= SQA \times SOG \times (1 - SQNSV) / GA_{<} \\ &= SQA \times SOG \times (1 - SQNSV) / (SOG_{<} \times (1 - SQNSV_{<})) \\ &= SQA \times (SOG / SOG_{<}) \times (1 - SQNSV) / (1 - SQNSV_{<}) \end{aligned}$$

where

$XX_{<}$ = League Average for statistic XX

Note $SQNSV_{<} = SV_{<}$

This says that you can express goals against as a percentage of average as the **product** of three indices:

$$\begin{aligned} \text{Goals Against Index} &= \text{Shot Quality Index} \times \text{Shots Allowed Index} \times \text{Goaltending Index} \\ GAI &= SQA \times SAI \times GI \end{aligned}$$

The first two indices can also be combined as a Defense Index. In each case, a lower value is a better value. Below is a table of these index values for 2003 and for 2004.

Team	2003				2004			
	Goals Against Index	Shot Quality Index	Shots Allowed Index	Goaltending Index	Goals Against Index	Shot Quality Index	Shots Allowed Index	Goaltending Index
NJD	0.763	0.915	0.831	1.003	0.779	0.915	0.870	0.978
DAL	0.776	0.972	0.891	0.896	0.831	0.998	0.827	1.007
CAL	1.047	0.953	0.955	1.151	0.836	1.013	0.909	0.908
MIN	0.818	0.947	1.004	0.860	0.869	0.946	1.036	0.886
SJ	1.098	1.022	1.028	1.045	0.869	0.970	1.032	0.869
PHI	0.763	0.935	0.868	0.940	0.883	0.928	0.909	1.046
BOS	1.089	1.027	0.998	1.062	0.893	0.991	0.991	0.908
DET	0.933	0.976	1.015	0.941	0.897	1.025	0.935	0.936
OTT	0.836	0.970	0.874	0.987	0.897	0.987	0.887	1.026
MON	1.075	0.982	1.151	0.950	0.912	0.927	1.019	0.965
TB	0.965	0.965	0.988	1.012	0.912	0.961	0.906	1.047
VAN	0.956	1.004	0.939	1.013	0.921	0.961	0.949	1.011
COL	0.891	1.018	0.999	0.877	0.940	0.983	1.011	0.947
STL	1.020	1.087	0.880	1.066	0.940	1.050	0.914	0.979
TOR	0.956	0.956	1.042	0.959	0.969	0.986	0.942	1.043
EDM	1.057	1.018	0.966	1.075	0.988	0.942	0.946	1.109
CAR	1.103	1.024	1.002	1.074	0.992	1.022	0.935	1.039
NYI	1.061	1.037	0.996	1.027	0.997	0.965	0.976	1.059
ANA	0.887	0.972	1.022	0.893	1.011	1.049	1.062	0.908
LA	1.015	1.048	0.920	1.053	1.030	1.072	0.962	0.999
NAS	0.946	1.000	0.968	0.978	1.030	0.985	1.015	1.031
BUF	1.006	0.969	0.987	1.052	1.049	0.939	1.017	1.099
FLA	1.089	1.078	1.171	0.862	1.049	1.000	1.229	0.853
CBJ	1.208	1.041	1.135	1.023	1.130	0.980	1.140	1.011
ATL	1.305	1.045	1.115	1.121	1.154	1.103	1.048	0.998
PHO	1.057	0.980	1.057	1.020	1.163	1.056	1.089	1.011
NYR	1.061	1.057	1.043	0.962	1.187	1.124	1.042	1.014
WAS	1.011	0.973	1.055	0.985	1.201	0.994	1.124	1.075
CHI	1.038	0.996	1.008	1.034	1.230	1.035	1.096	1.084
PIT	1.172	0.996	1.090	1.079	1.439	1.043	1.182	1.166

Here are a few random, closing observations:

- **New Jersey again led the NHL in goal prevention**, but moved from 76.3% of league average goals to 77.9%. The Devils led in SQA. There was a deterioration in the shots index (still a rank of 2nd), but an improvement in goal.
- **The Bruins were a very average defensive team** (SQA .991, SAI .991) until you got to the crease (GI .908). They rode Raycroft to improved goal prevention, rising from 24th to 7th.
- **Montreal had a huge improvement in team defense** (DI from 1.130 to .945) with the same goaltending. Toronto and Montreal had similar numbers before consideration of shot quality. Toronto had an SAI of .942 and a GI of 1.043. Montreal had an SAI of 1.019 and a GI of .965. For both teams, when you multiply the SAI by the GI you get .983. Montreal allowed 192 goals against (8 fewer than Toronto). **Montreal's superior shot quality was therefore worth about 3 points in the standings.**
- The Penguins and the Panthers were the two worst defensive teams in the NHL. The only difference was in goal. Although Florida's shot quality improved to 1.000, the shots allowed index was still 1.229. Of course, Luongo rescued this with a GI of .853. Pittsburgh (DI 1.233) allowed fewer shots (SAI 1.182) but shot quality (SQA 1.043) was worse and the net effect was about the same. But they poured gasoline on the fire in goal (GI 1.166).
- Although they got there in very different ways, Florida and Edmonton had similar numbers before consideration of shot quality. Florida had an SAI of 1.229 and a GI of .853. Edmonton had an SAI of .946 and a GI of 1.109. For both teams, when you multiply the SAI by the GI you get 1.048/9. Edmonton allowed 208 goals against. The Panthers allowed 13 more goals. **Edmonton's superior shot quality was therefore worth about 5 points in the standings.**
- St. Louis (SAI .914 x GI .979 = .895) and San Jose (1.032 x .869 = .897) also had similar numbers before consideration of shot quality. San Jose allowed 183 goals against while St. Louis allowed 198. **San Jose's superior shot quality was therefore worth about 6 points in the standings.**
- Tampa Bay considerably reduced their shots on goal. **Did you ever hear that the best defense is a good offense? Well it seems to be somewhat true with shots on goal.** Tampa Bay had the third most improved offense in 2004. Along the way, their shots allowed improved to 4th. The other two teams with big improvements in offense (Buffalo and Nashville) did not show this defensive improvement, however, in 2004, the correlation coefficient between Goals For and Shots Allowed was -49%. The negative sign says that when goals scored go up, shots allowed go down. The "49%" says that about 24% (.49²) of the variation in Shots Allowed is attributable to variation in Goals For. Statistically, this is a weak but significant relationship. More synergies in hockey ...